ICE FCOJ-A FUTURES (OJ)

"NYBOT Orange Juice" -- Spread & Curve Structure Fundamental Reference

Florida Freeze Risk, Brazilian Supply, and Citrus Greening Disease

This document presents a fact-based reference on the fundamental drivers of ICE Frozen Concentrated Orange Juice futures (ticker: OJ, formally "FCOJ-A"), traded on ICE Futures US and still commonly referred to by market participants as "NYBOT Orange Juice" after its originating exchange -- the contract has traded in New York since 1966, first on the New York Cotton Exchange, then its successor the New York Board of Trade (NYBOT), and now on ICE Futures US following ICE's 2007 acquisition of NYBOT. FCOJ-A is, by both contract design and market structure, one of the most distinctive markets covered in this document series: it is the world benchmark for a single processed citrus product whose underlying supply is concentrated in two specific growing regions -- Florida and the state of Sao Paulo, Brazil -- and whose price history has been shaped to an unusual degree by a small number of acute weather and disease events. Florida freeze risk and US Atlantic hurricane risk give FCOJ-A a documented asymmetric, spike-prone price distribution that industry literature has described as functionally similar to owning a call option on a supply disruption, while the multi-decade spread of citrus greening disease (HLB) across both Florida and, more recently, Brazilian growing regions has become the dominant structural supply story of the past fifteen years. This document presents the FCOJ-specific fundamentals in full, with particular emphasis on the Florida and Brazilian growing regions, the citrus greening disease crisis, weather risk mechanics, and the structural decline in orange juice consumption that has reshaped demand over the contract's recent history.

Futures First -- Internal Education Material
For Analyst Group Use

1. Contract Specifications and Delivery Mechanics

FCOJ-A is a physically delivered contract for US Grade A frozen concentrated orange juice, with a defined list of eligible countries of origin reflecting the small number of regions capable of supplying juice meeting the contract's grade and concentration specification at commercial scale.

Parameter	Detail
Exchange	ICE Futures US (formerly NYBOT, itself successor to the New York Cotton Exchange)
Ticker	OJ (front month); formally "FCOJ-A"; commonly called "NYBOT Orange Juice" or simply "OJ"
Underlying	US Grade A frozen concentrated orange juice (FCOJ), not less than 62.5 degrees Brix, graded by USDA
Contract size	15,000 pounds of orange solids
Quotation	US cents per pound (¢/lb); minimum tick = 0.05¢/lb = $7.50 per contract
Contract months	January (F), March (H), May (K), July (N), September (U), November (X) -- six delivery months per year. Contracts are not recognised by the Exchange extending beyond 36 months forward.
Settlement type	Physical delivery by tank, requiring an Exchange Warehouse Receipt (EWR) registered for FCOJ meeting tank-delivery specifications, at Exchange-licensed warehouses at several US delivery points.
Eligible countries of origin	United States, Brazil, Costa Rica, and Mexico -- a notably short list reflecting the limited number of growing regions capable of producing juice meeting the contract grade at commercial volume.
Delivery grade detail	Bottom (sinking) pulp limited to a maximum of 12% on initial test; recoverable oil between 0.005% and 0.020%; FCOJ above 66 degrees Brix calculated at 7.278 lb of solids per gallon delivered.
Price limits	Subject to a daily price limit that can range from 10 to 30 cents per pound depending on the prevailing Designated Price Limit (DPL) level set by the Exchange -- a wider potential daily limit range than most other soft commodity contracts in this document series, reflecting the market's documented history of sharp, weather-driven price moves (Section 5).
Last trading day	The Exchange-specified final trading day in the delivery month; first notice day is the first business day of the expiring delivery month.
Contract value at $2.50/lb	$37,500 per contract

1.1 FCOJ vs Not-From-Concentrate (NFC) -- Why the Futures Benchmark Persists

- **FCOJ's declining share of the physical retail market**: not-from-concentrate (NFC) orange juice -- juice that is squeezed, pasteurised, and chilled without the freeze-concentration step -- surpassed FCOJ in US retail market share during the 1980s and has remained the dominant retail product form since, reflecting consumer preference for what is marketed as a fresher product.
- **Why FCOJ remains the pricing benchmark despite this**: FCOJ's storability and ease of shipping (it can be frozen, stored for extended periods, and transported efficiently in concentrated form before being reconstituted) make it well suited to function as a standardised, exchange-tradeable commodity in a way NFC juice -- which must be kept chilled and has a much shorter shelf life -- is not. The industry has therefore continued to use the FCOJ futures market as the primary price discovery

and hedging mechanism for the broader orange juice complex, including NFC juice, even as FCOJ's own direct share of retail consumption has declined.

2. Global Production -- Florida and Brazil as the Two Pillars

World orange juice supply for the FCOJ/NFC complex is concentrated overwhelmingly in two growing regions -- the US state of Florida and the Brazilian state of Sao Paulo -- to a degree of geographic concentration unusual even among the soft commodities covered elsewhere in this document series.

Region	Role	Harvest/Crop Calendar	Key Structural Facts	OJ Curve Relevance
Brazil (Sao Paulo state)	The world's largest orange juice producer and exporter by a wide margin	Marketing year aligned with the Brazilian agricultural calendar, harvest concentrated roughly June-January depending on variety	Highly concentrated, large-scale, vertically integrated processing industry; a small number of major processors account for the substantial majority of Brazilian FCOJ export volume	Brazilian crop estimates (published by Fundecitrus, Section 6) are the single most important data series for the global balance
United States (Florida)	The historically dominant US producer, now substantially diminished by citrus greening disease (Section 4)	US marketing year begins December 1; Florida harvest runs roughly October through June depending on variety (early/midseason vs Valencia)	Florida orange production has fallen to multi-decade-low levels (documented in Section 4); the 2022/23 Florida crop was the smallest since 1936, and the 2023 total US crop the lowest in 86 years	Despite its reduced absolute volume, Florida weather risk (freeze and hurricane, Section 5) remains the dominant acute price-spike driver given the contract's US delivery basis
Mexico	A smaller producer and an eligible delivery origin	Varies by region	A secondary supply source; included among the contract's eligible countries of origin	A minor but eligible contributor to deliverable supply
Costa Rica	A smaller producer and an eligible delivery origin	Varies	A secondary supply source; included among the contract's eligible countries of origin	A minor but eligible contributor to deliverable supply

2.1 The US Marketing Year and Crop Reporting

- The US orange marketing year begins December 1 (e.g., the 2026/27 marketing year runs December 1, 2026 through November 30, 2027), aligned with the Florida harvest calendar, and is the convention used by USDA and most US trade statistics for the orange and FCOJ balance sheet.

3. Demand Fundamentals -- A Structurally Declining Consumption Base

Unlike most of the agricultural commodities covered elsewhere in this document series, orange juice demand in its largest single market (the United States) has been in well-documented, multi-decade structural decline -- a demand-side feature that distinguishes FCOJ's long-run fundamental picture from most other contracts in this series.

3.1 The US Consumption Decline

- **Documented multi-year consumption decline**: US orange juice consumption has fallen substantially over recent decades, with USDA data showing US consumption reaching a 5-year low in 2023 even before accounting for the longer-run decline trend, reflecting shifting consumer breakfast habits, competition from other beverage categories, and -- particularly in recent years -- some health-conscious consumer movement away from sugar-containing beverages including fruit juice.
- **The interaction between record-high prices and demand destruction**: the historic 2023 price spike (Section 5.3) illustrated this demand-side feature directly: as FCOJ prices reached record highs amid the severe supply shortfall, USDA reported that the resulting retail price increases began to curb consumer demand, with US orange juice consumption falling to a multi-year low even as the immediate supply shock was the primary driver of the price move -- demonstrating that, unlike commodities with more inelastic demand, sustained extreme pricing can itself become a self-correcting demand-side factor for orange juice specifically.

3.2 Cold Storage Stocks

- **USDA cold storage data**: published periodically, tracking US FCOJ inventories held in cold storage -- a direct read on the balance between supply (Florida and imported Brazilian FCOJ) and consumption. The widely reported decline in US FCOJ cold storage stocks to a 54-year low in December 2023 was a key contemporaneous data point confirming the severity of the supply shortfall driving that year's record prices.

4. Citrus Greening Disease (HLB) -- The Dominant Structural Crisis

Citrus greening disease, known scientifically as Huanglongbing (HLB) and colloquially as "citrus greening," is the single most important structural fact shaping FCOJ fundamentals over the past fifteen years, having reduced Florida's orange production capacity more severely than any weather event in the contract's history.

4.1 What Citrus Greening Is

- **Disease mechanism**: HLB is a bacterial disease, transmitted by the Asian citrus psyllid insect, that infects the tree's vascular system, progressively reducing fruit yield and quality and ultimately killing the tree over a multi-year period. There is no commercially available cure once a tree is infected, and the disease has no precedent for natural recovery.
- **Why it is more severe than a weather event**: unlike a frost or hurricane, which damages a single season's crop or, in severe cases, kills trees outright in an acute event, HLB operates as a slow, compounding, multi-year reduction in productive capacity across the entire affected growing region -- meaning its effect on the long-run supply trajectory has been more severe and more difficult to reverse than any single acute weather event covered in Section 5.

4.2 Florida's HLB Crisis

- **Documented production collapse**: Florida orange production has fallen to levels not seen in many decades as a direct consequence of HLB's spread through Florida groves since the disease was first confirmed in the state in the mid-2000s.
- **Compounding with acute weather events**: the disease's progressive weakening of Florida's tree stock has, on several documented occasions, compounded with acute hurricane and freeze damage (Section 5) to produce historically severe crop shortfalls -- since HLB-weakened trees are, per industry and university extension research, generally less resilient to additional weather stress than healthy trees, and storm damage further slows the grove-renewal and replanting efforts that represent the primary long-run response to HLB.

4.3 Citrus Greening's More Recent Spread to Brazil

- HLB has also been confirmed and has spread within Brazilian growing regions, a development of particular significance for the global FCOJ balance given Brazil's position as the world's dominant producer (Section 2) -- meaning the disease is no longer a Florida-specific structural risk but a developing concern for the world's primary supply source as well, a structural risk trajectory that market participants and industry analysts continue to monitor as it develops, since the pace and ultimate scale of Brazilian HLB impact remains a subject of ongoing assessment rather than a fully resolved historical fact.

4.4 Grove Renewal and Replanting Efforts

- Florida's citrus industry, supported by university extension research (notably from the University of Florida's Citrus Research and Education Center, established in 1948 and a long-standing centre of FCOJ-related agronomic research) and various state and federal programmes, has pursued grove renewal using HLB-tolerant rootstock and tree varieties, though the multi-year time lag between planting and full commercial production (citrus trees typically require several years to reach significant bearing capacity) means any meaningful recovery in Florida production operates on a long structural timeline.

5. Weather Risk -- Florida Freeze and Hurricane Mechanics

FCOJ's documented history of sharp, asymmetric price spikes is rooted substantially in two specific Florida weather risk windows, each operating through a distinct physical mechanism.

5.1 The Florida Freeze Season

- **Timing**: late November through March, the period during which Florida -- despite its generally subtropical climate -- remains exposed to occasional polar air incursions capable of producing freezing or near-freezing temperatures in citrus-growing regions.
- **Mechanism and severity gradient**: a freeze event's effect on the crop depends on temperature, duration, and whether the cold occurs alongside still air or wind (radiational versus advective freezes, in agronomic terminology), with the most severe events capable of damaging or destroying the standing crop and, in the most extreme historical cases, killing trees outright -- a tree-killing freeze event compounds into a multi-year supply disruption analogous in concept to the Brazilian coffee frost mechanism described in the Coffee C reference document elsewhere in this series, though operating on a citrus tree rather than a coffee tree and within Florida's specific geography.
- **Why this remains a live risk despite Florida's reduced crop size**: even with Florida's HLB-diminished production base (Section 4.2), the remaining Florida crop continues to carry freeze exposure, and a severe freeze event affecting an already HLB-weakened, reduced grove base can still produce a historically significant percentage crop loss, even if the absolute tonnage at risk is smaller than in past decades.

5.2 US Atlantic Hurricane Season

- **Timing**: officially June 1 through November 30, overlapping with a portion of the Florida growing season ahead of harvest.
- **Damage mechanisms**: direct wind damage knocking fruit from trees prematurely (a particular risk closer to harvest timing, when fruit is larger and more vulnerable to wind-driven drop), flooding and waterlogged soil affecting tree health, and, in the most severe cases, structural damage to trees themselves -- with the specific severity and crop-year impact depending heavily on a storm's track relative to the concentrated citrus-growing counties in central and southern Florida.
- **Documented historical example**: Hurricane Irma's 2017 impact on Florida groves was a documented, market-moving event, contributing to the USDA's subsequent cut to a 72-year-low Florida crop estimate for that season -- illustrating the hurricane-to-crop-estimate-revision mechanism that the market monitors closely during each Atlantic hurricane season.

5.3 The 2023 Price Spike -- A Documented Illustration of Compounding Risk

- The historic 2023 FCOJ price spike to a record 431.95 cents in October illustrates the compounding-risk framework described throughout this section: the 2022/23 Florida crop, already severely reduced by the ongoing HLB crisis (Section 4.2) to its smallest level since 1936, was further affected by storm damage during the relevant hurricane season, while simultaneously Brazilian inventories -- the supply the market would normally lean on to offset a Florida shortfall -- had fallen to a record low, removing the usual buffer and allowing the combined shortfall to drive prices to their highest level in the contract's history, with the subsequent demand destruction described in Section 3.1 only partially moderating the price move into year-end.

6. Brazilian Crop Data and the Global Balance

6.1 Fundecitrus -- The Primary Brazilian Data Source

- **Fundecitrus (Fundo de Defesa da Citricultura)**: a Brazilian citrus industry research and crop-estimation organisation, publishing the primary Brazilian orange crop estimates used by the global trade, analogous in market function (though not institutional structure) to CONAB's role for Brazilian soybean and corn estimates described elsewhere in this document series. Fundecitrus crop estimates, published periodically through the Brazilian growing season, are among the most closely watched data releases for the global FCOJ balance.

6.2 USDA Foreign Agricultural Service and World Markets Reports

- USDA FAS publishes periodic citrus: world markets and trade reports covering global orange and orange juice production, consumption, and trade, providing a comprehensive cross-check against Fundecitrus's Brazil-specific data and USDA NASS's US-specific data.

6.3 Processor Concentration in Brazil

- The Brazilian FCOJ export industry is characterised by a small number of large, vertically integrated processors controlling a substantial share of Sao Paulo orange processing and export capacity -- a market structure feature that means individual processor-level operational news (processing capacity utilisation, inventory policy) can carry outsized significance for the global FCOJ balance relative to a more fragmented producer base.

7. The Forward Curve Structure

FCOJ trades in six delivery months per year (January, March, May, July, September, November), with the curve shape reflecting the combined influence of the US marketing year/Florida harvest calendar, the Florida freeze and hurricane risk windows, and the Brazilian crop cycle, which operates on a different calendar to the US.

7.1 Carry, Backwardation, and the Weather Premium

- **Normal carry**: in a well-supplied market, FCOJ trades in a modest contango reflecting storage, financing, and insurance costs for holding frozen concentrate -- a product that, unlike NFC juice, stores well over an extended period given proper frozen storage.
- **Backwardation as a tightness signal**: as with the other soft commodities in this series, FCOJ backwardation signals near-term physical scarcity -- low cold storage stocks (Section 3.2), confirmed Florida crop damage, or a Brazilian inventory shortfall of the kind documented in the 2023 episode (Section 5.3).
- **The embedded weather-risk premium**: contracts spanning the Florida freeze season (roughly the January and March contracts) and the hurricane season (roughly the July, September, and November contracts) can carry an embedded weather-risk premium relative to a pure-carry baseline, reflecting the market's assessment of acute weather probability during the relevant window -- a curve phenomenon directly analogous to the Brazilian coffee frost premium described in the Coffee C reference document, here operating across two distinct seasonal windows (freeze and hurricane) rather than coffee's single winter frost window.

8. Macro and Cross-Market Drivers

8.1 Brazilian Real (BRL/USD)

- As with other Brazilian-sourced agricultural exports described throughout this document series (sugar, coffee, soybean meal), BRL/USD movement affects Brazilian orange grower and processor economics through the same general mechanism: BRL depreciation improves the BRL-equivalent revenue Brazilian exporters receive at a given USD FCOJ price, incentivising export volume and selling pace, while BRL appreciation has the opposite effect.

8.2 Speculative Positioning

- **CFTC COT for FCOJ**: weekly managed money net positioning is tracked, consistent with the positioning framework described throughout this document series. FCOJ's documented history of sharp, weather-driven price spikes (Section 5) has, per industry literature, made it a market of particular interest to commodity trading advisors and funds specifically for its diversifying properties relative to other asset classes and its asymmetric, option-like payoff profile during acute supply disruption episodes.

9. Documented Seasonal Patterns and Historical Events

9.1 Seasonal Patterns

- **The Florida freeze season (late November-March)**: the most acute, calendar-certain weather-risk window of the year, during which market attention to Florida temperature forecasts rises sharply regardless of the specific forecast at any given moment.
- **US Atlantic hurricane season (June 1-November 30)**: a second major weather-risk window, with the specific threat to Florida citrus depending on storm track and timing relative to harvest.
- **Brazilian harvest progress (roughly June-January)**: Fundecitrus estimates and harvest progress reports through this period provide the primary read on the dominant global supply source.

9.2 Documented Historical Events

- **Hurricane Irma (2017)**: documented storm damage to Florida groves contributing to a subsequent USDA crop estimate cut to a 72-year low for that season, as referenced in Section 5.2.
- **The citrus greening (HLB) crisis (mid-2000s-ongoing)**: the dominant multi-year structural story for Florida production, described in full in Section 4, with production reaching successive multi-decade lows through the 2020s.
- **The 2022/23-2023 compounding supply crisis**: as detailed in Section 5.3, the combination of the smallest Florida crop since 1936, storm damage, and record-low Brazilian inventories drove FCOJ to a record nominal high of 431.95 cents in October 2023, with US cold storage stocks simultaneously falling to a 54-year low -- the most significant documented price episode in the contract's recent history, followed by demand destruction (Section 3.1) as a partial market-correcting response.

10. Key Data Sources and Release Schedule

Report / Data	Publisher	Frequency	Typical Timing	Primary OJ Curve Impact
USDA NASS Florida/US Citrus Crop Estimates	USDA NASS	Periodic through the season	Variable	All contracts -- the primary official US crop size data series
Fundecitrus Brazilian crop estimates	Fundecitrus (Brazil)	Periodic through the Brazilian season	Variable	All contracts -- the primary data series for the dominant global supply source (Section 6.1)
USDA FAS Citrus: World Markets and Trade	USDA Foreign Agricultural Service	Periodic	Variable	All contracts -- comprehensive global produ ction/consumption/trade cross-check
USDA Cold Storage report (FCOJ component)	USDA NASS	Periodic, monthl y-equivalent cadence	Variable	Front contracts -- US inventory tightness signal (Section 3.2)
NOAA/NWS Florida temperature forecasts and freeze warnings	NOAA National Weather Service	Daily during freeze season	Continuous	Front-to-mid contracts, acutely during Nov-Mar -- the single fastest-moving signal during freeze risk windows
NOAA National Hurricane Center storm tracking	NOAA NHC	Continuous during hurricane season	Continuous	Mid-curve contracts spanning Jun-Nov -- storm track relative to Florida citrus counties
BRL/USD exchange rate	Reuters / Bloomberg / BCB	Daily/Continuous	Market hours	All contracts -- Brazilian grower/exporter selling economics
CFTC Commitments of Traders (FCOJ)	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding -- all contracts

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ICE FCOJ-A is, among the soft commodity contracts covered in this document series, distinguished by the narrowness of its global supply base -- two growing regions, Florida and Sao Paulo, dominate world supply to a degree with few parallels elsewhere in this series -- and by the layering of an acute, calendar-bound weather risk (Florida freeze and hurricane seasons) on top of a slow-moving but severe structural disease crisis (citrus greening/HLB) that has fundamentally reduced Florida's productive capacity over the past fifteen years and is now an emerging concern for the Brazilian supply base as well. Layered on top of this unusually concentrated and risk-exposed supply picture is a demand base in well-documented multi-decade structural decline in the contract's largest consuming market, giving FCOJ a combination of supply fragility and demand erosion with few direct parallels among the commodities covered elsewhere in this series.

For spread traders, the most reliable framework tracks the two Florida weather-risk windows (freeze, late November-March; hurricane, June-November) as calendar-certain volatility periods regardless of any specific forecast, Fundecitrus Brazilian crop data as the primary read on the dominant global supply source, cold storage stocks as the key US physical-tightness signal, and the ongoing citrus greening trajectory in both Florida and Brazil as the structural backdrop against which any acute weather event must be assessed -- since, as the 2023 episode demonstrated, a weather or logistics shock landing on an

already HLB-diminished supply base produces a materially larger price reaction than the same shock would against a healthy, ample supply base.